

2Market

Data analysis for marketing and sales

1 Introduction

2Market is a supermarket interested in understanding the purchase behaviour of their customer. Therefore, an analysis of their data is conducted, to correlate customer details with their preferred product.

In this document, the outcomes of the data analysis are given, and possible marketing strategies are suggested.

2 Analytical Approach

Two data files were given, one with marketing data, another with the advertisement data. Both were cleaned with Excel.

Missing values, empty cells, spelling errors and duplicates were searched, as well as #NULL! Or #Value, #DIV/0#REF errors.

The marketing data file presented some formatting issues on the Education and Marital status columns. Decimal places were checked, but no errors were spotted. Finally, the date column was fixed with the TEXTPLIT command.

The data were then used to remove the outliers in the Age column. The age range was chosen by calculating the first and third quartiles (Q1 and Q3, respectively), the interquartile range (IQR), and the upper and lower limits, Table 1.

Table 1 - Calculated properties for the age range.

	Age
Q1	46
Q3	64
IQR	18
Upper Limit	91
Lower Limit	19

Some of questions about customer demographics were answered.

The average age of 2Market's customers is 54. To calculate the average, an array of ages is, called Age, with the help of the year_birth (column B). For example, the age of B2 cell =YEAR(TODAY())-B2. Then, in cell Y2 we can calculate the average as =AVERAGE(X2:X2217)

For each marital status, the average age can be calculated with the AVERAGEIF command. For example, the formula AVERAGEIF(D2:D2217,"Absurd",X2:X2217) is used to calculated the average age of the absurd marital status.

Other groups can be filtered and analysed, with the help of pivot tables. For example, the average age of the group composed of married PhD customers with two teens at home is 63,

Table 2.

Table 2 - Average age of married PhD customers with two teens at home

Kidhome	(All)	
Marital_Status	Married	
Education	PhD	
Teenhome	2	
Average of Age		
	63	

The average age of customers who earn a yearly income between \$90K and \$100K is 52. This calculation can be performed by filtering the data. Select all the table, then Home → Editing → Sort & Filter. Then, click on the arrow in the Income (column E), select Number Filters → Between → Show results where Income is greater than or equal than 90000 And is less than or equal than 100000. Then, the average can be calculated as =SUM(X2164:X2204)/COUNT(X2164:X2204). The formula will use only the selected cells.

The average age by marital status can be calculated from the Pivot table, and a histogram can be plotted, Figure 1.

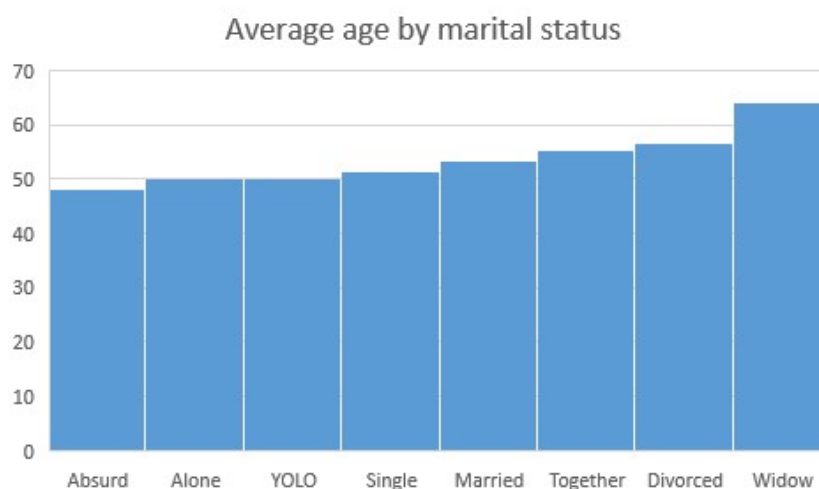


Figure 1 - Average age by marital status

The highest average age belongs to the widow marital status (64), while the lowest average age belongs to the absurd marital status (48).

To check for trends, anomalies and outliers, a box and whisker plot can be created, Figure 2.

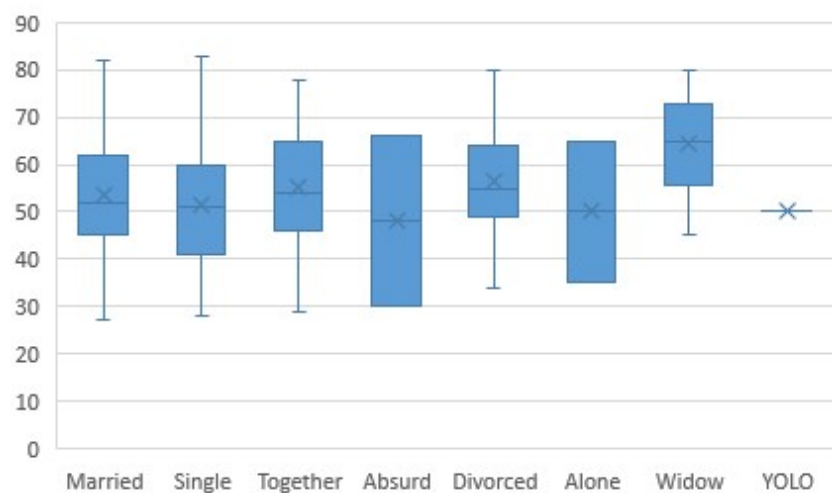


Figure 2 – Box and whisker for age by marital status

The highest average age (X signs in boxes) by married status is Widow, the lowest average age is the Single, but with a wide spread of ages.

Absurd and YOLO marital status have only two entries, so no variations are present. Besides, YOLO entries have the same value, so no bar either. Outliers are present among the divorce, single and together marital status, with an age between 120 and 140 years old.

The average age decreases from the Divorced, Married, and Single marital status, then it increases from Together and Widow. Alone, YOLO and Absurd have a similar average age than Single.

Besides, the average income increases with increasing of age, up to approx. \$60k, then it slowly decreases, Figure 3.

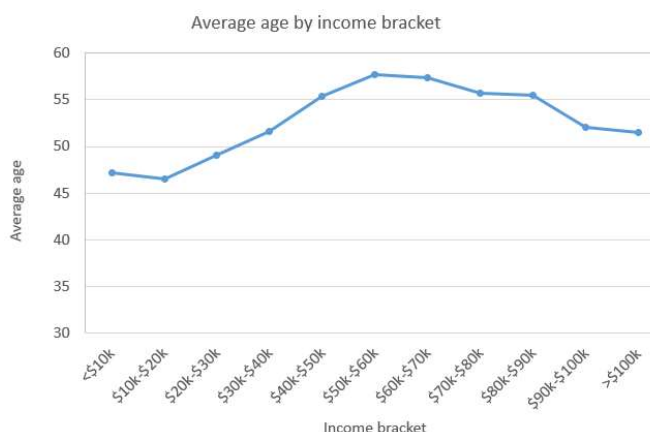


Figure 3 – Average age by income bracket

3 Customer demographic and sales information

In order to deliver insights on customer demographic and sales information, a Tableau dashboard is drawn, Figure 4.

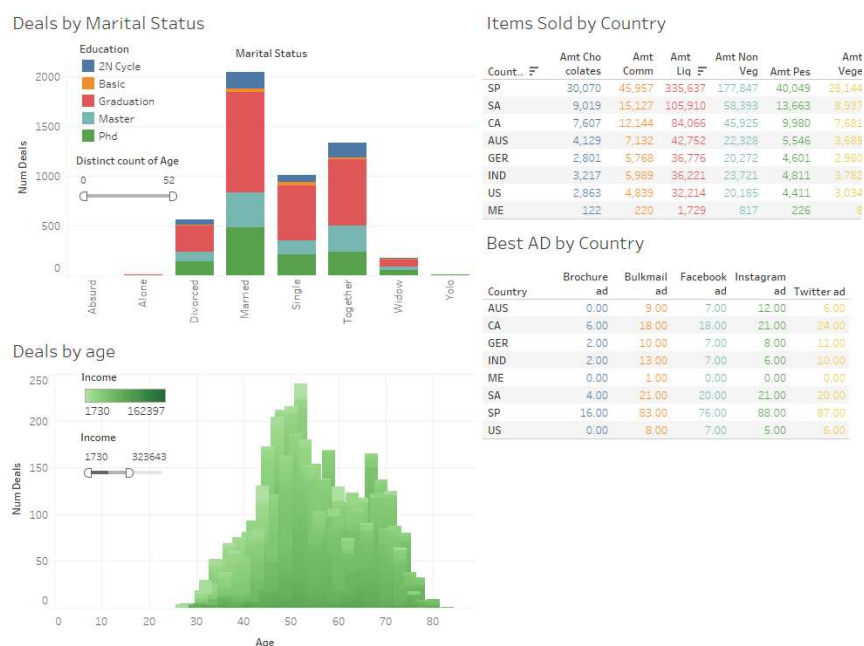


Figure 4 – Tableau dashboard

Here, number of deals are ordered by marital status and age. Besides, insights on items sold by country and the best AD per country are given.

The dashboard is divided in four areas, i.e., deals per marital status on top left corner, items sold by country on top right corner, deals by age on bottom left corner, best AD by country on bottom right corner.

1. Deals by marital status.

A histogram on number of deals ordered by marital status is shown. Each bin is divided by education. The age range can be changed with the slide. The highest spending marital status is married, followed by together, single, divorced, and widow. Graduated customers are the highest group in the alone, divorced, married, single, together, and widow groups.

2. Items sold by country.

In the top right corner, a table with number of items sold by country is shown. Here it can be noticed that Spain is the top country by number of items sold.

3. Deals by age.

In the bottom left corner, a histogram on the number of deals by age is shown. The income is colour-shaded, from light green for small incomes, to dark green for high incomes. The income range can be selected with the slide. People around 50 years old close the highest number of deals.

4. Best AD by country.

In the bottom right corner, a table showing the number of AD per country is shown. Here, it can be noticed that social media ads are the most successful ones, in all the countries.

4 Data Query (SQL)

Both the files were imported in an SQL database and queried for further demographic analysis.

The total spend per country was calculate by summing the amount spent per good and grouped by country, Table 3.

Table 3 - Total amount spend per country

	country character varying (10)	TotalSpendPerCountry bigint
1	SP	659557
2	SA	211071
3	CA	167403
4	AUS	85576
5	IND	77806
6	GER	73198
7	US	67546
8	ME	3122

Similarly, the total spend per product and per country was calculated by summing each product category, and grouped by country, Table 4.

Table 4 - total amount spent per good, per country

	country character varying (10)	total_liq bigint	total_vege bigint	total_non_veg bigint	total_pes bigint	total_chocolates bigint	total_comm bigint
1	SP	336392	28288	178409	40153	30134	46181
2	CA	84066	7681	45925	9980	7607	12144
3	IND	36236	3788	23729	4818	3221	6014
4	AUS	42752	3689	22328	5546	4129	7132
5	US	32214	3034	20185	4411	2863	4839
6	ME	1729	8	817	226	122	220
7	SA	105918	8937	58398	13670	9019	15129
8	GER	36776	2980	20272	4601	2801	5768

The most popular product per country was calculated by checking which product category was the most popular, by WHEN GREATEST SUM commands for each product category, and grouped by country. Doing a similar calculation, but grouping the results by marital status, gave the most popular product by marital status, Table 5.

Table 5 – Most popular products by marital status

	marital_status character varying (50)	most_popular_product text	product_sale_amount bigint
1	Widow	Liquor	27902
2	YOLO	Liquor	644
3	Together	Liquor	176715
4	Alone	Liquor	554
5	Absurd	Liquor	711
6	Married	Liquor	256976
7	Single	Liquor	137217
8	Divorced	Liquor	75364

All the marital status prefer alcohol.

5 Conclusions

From this analysis, it can be concluded that 2Market has the best possibilities to increase sales if the future marketing campaign will be focus on alcohol products for married and graduated Spanish people, by using Instagram.